

## Conductive Ceramic Materials

### types of conductivity

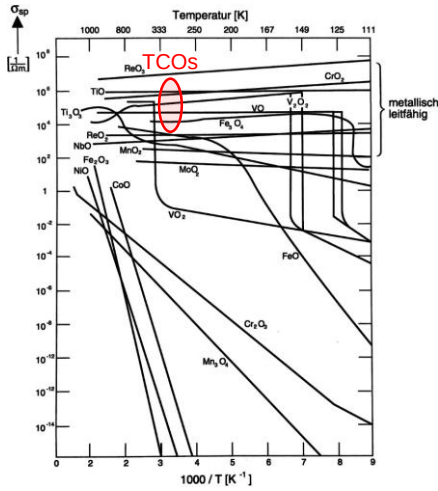
#### conductors of the 1<sup>st</sup> type:

- are not materially changed during conduction;
- applies to all metals and semiconductors, but also to some ceramic materials (e.g. TCO materials)
- conductivity through the transport of electrons  $e^-$  (or holes  $h^+$ )

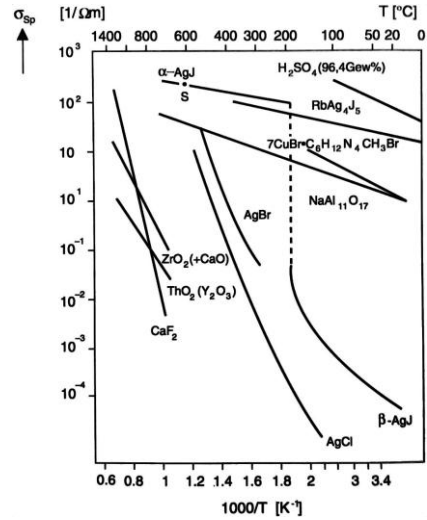
#### conductors of the 2<sup>nd</sup> kind:

- are materially changed during conduction;
- applies to ionic conductors, which dissociate into mobile ions and the solid electrolyte;
- examples: salt solutions, ceramic ionic conductors with solid electrolyte, e.g. Y stabilized  $ZrO_2$   
→ conductivity due to the transport of ions  $M^{n+}$  or  $X^{n-}$

## electrically conductive ceramics

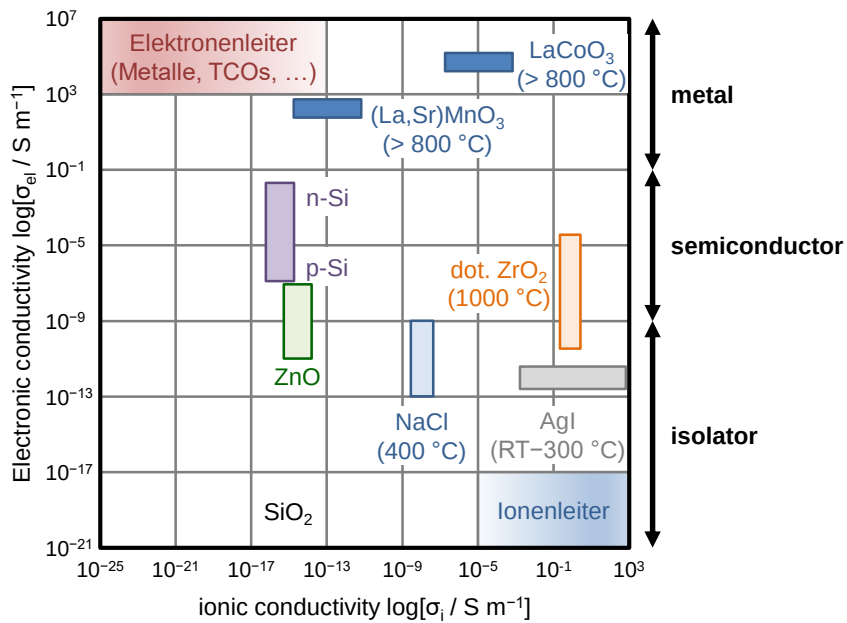


**1st kind:** electron conductor

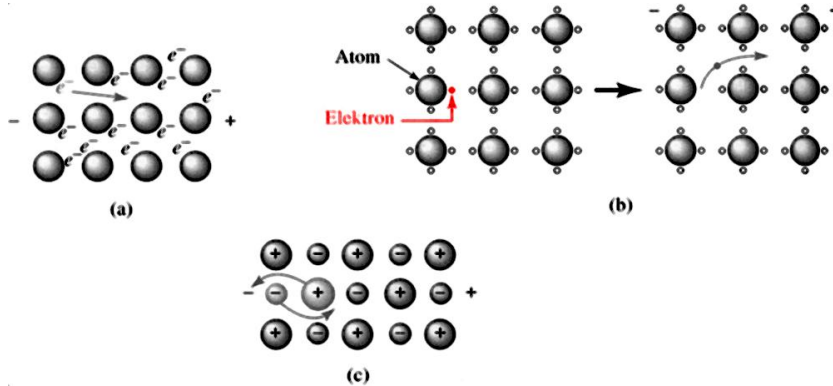


**2nd kind:** ion conductor; Conductivity 2....3 orders of magnitude lower compared to 1st kind

## electronic conductivity vs. ionic conductivity

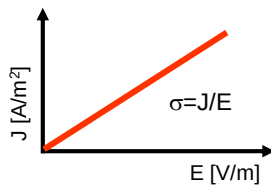


### types of charge carrier movement in solids (schematic)



- transport of (valence) electrons (in metals)  $\rightarrow$  simple;  $e^-$  readily available
- transport of electrons in semiconductors and insulator materials  $\rightarrow$  Bonds must first be broken, more difficult than a)
- transport of ions through solids  $\rightarrow$  diffusion-controlled (slow at room temperature)

### electric conductivity in ceramics



#### Ohmic (linear) behavior

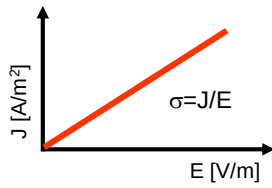
$J$  = current density

$E$  = el. field strength



conductivity/resistance determined  
by bulk properties: Bulk resistance  $R$ ,  
or conductance  $G$

## electric conductivity in ceramics



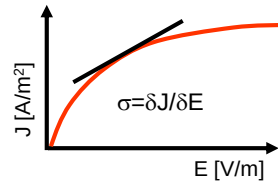
**Ohmic (linear) behavior**

$J$  = current density

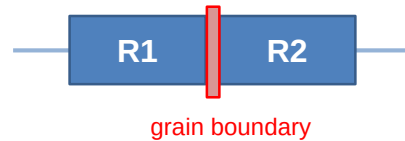
$E$  = el. field strength



conductivity/resistance determined by bulk properties: Bulk resistance  $R$ , or conductance  $G$

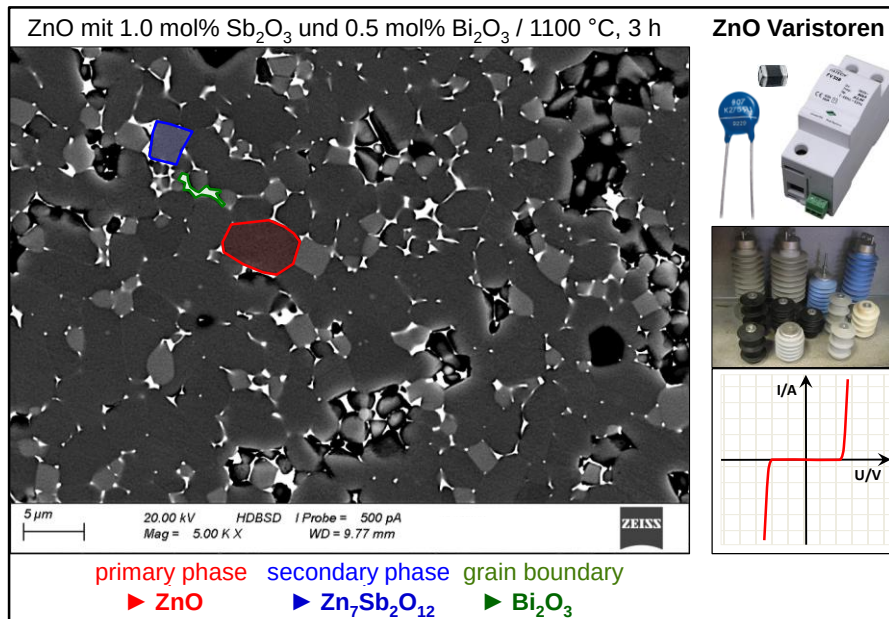


**nonlinear behavior**

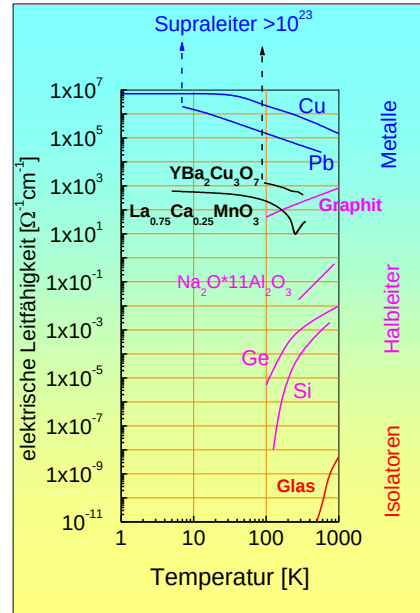
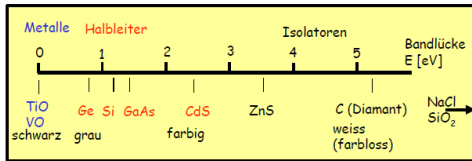
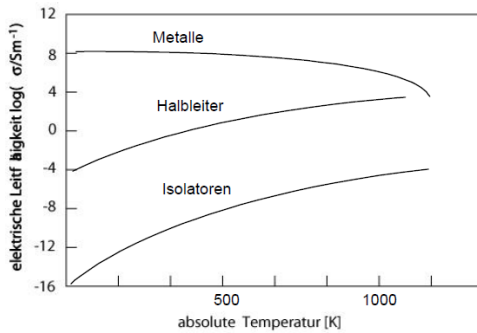


grain boundary

conductivity/resistance influenced by interfaces, grain boundary can act as barrier layer, above the breakdown voltage ohmic conductivity, below insulating  
→ example: Varistor ceramics (ZnO)



## temperature dependence of the conductivity in metals, semiconductors and insulators



What is current macroscopically?

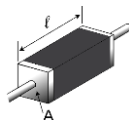
Ohm's law:

Linear relationship between  $U$  and  $I$  when a voltage flows through an electrical conductor through an electrical conductor:

$$U \sim I \rightarrow R = \frac{U}{I} \quad \text{or} \quad U = I \cdot R$$

adding the conductor geometry

$$R = \rho \cdot \frac{l}{A}$$



$\rho$  – spez. el. resistance /  $\Omega \text{ cm}$

$l$  – length of the conductor /  $\text{cm}$

$A$  – cross section of the conductor /  $\text{cm}^2$

due to the electrical resistance, heat is generated as power loss  $P$ :

or with

$$P = U \cdot I$$

$P$  – power loss /  $\text{W}$

$$U = I \cdot R$$

$$P = I^2 \cdot R$$

## Other forms of Ohm's law

$$R = \frac{U}{I} = \rho \cdot \frac{l}{A} \rightarrow \frac{I}{A} = \sigma \cdot \frac{U}{l}$$

$I/A$  – current density  $J$  /  $A \cdot cm^{-1}$

$U/l$  – electr. field strength  $\vec{E}$  /  $V \cdot cm^{-1}$

$\sigma$  – el. conductivity /  $\Omega^{-1} \cdot cm^{-1} = 1/\rho$

then the following applies to the current density  $J$ :

$$\vec{J} = \sigma \cdot \vec{E}$$

also applies to  $J$ :

$$\vec{J} = n \cdot z \cdot q \cdot v$$

$n$  – number of charge carriers per  $cm^3$

$z$  – charge number (for electron conductors = 1)

$q$  – elementary charge:  $1,602 \cdot 10^{-19} \text{ C}$

$v$  – average drift velocity/  $cm \cdot s^{-1}$

it follows:

$$\sigma \cdot \vec{E} = n \cdot z \cdot q \cdot v \quad \text{resp.} \quad \sigma = n \cdot z \cdot q \cdot \frac{v}{E}$$

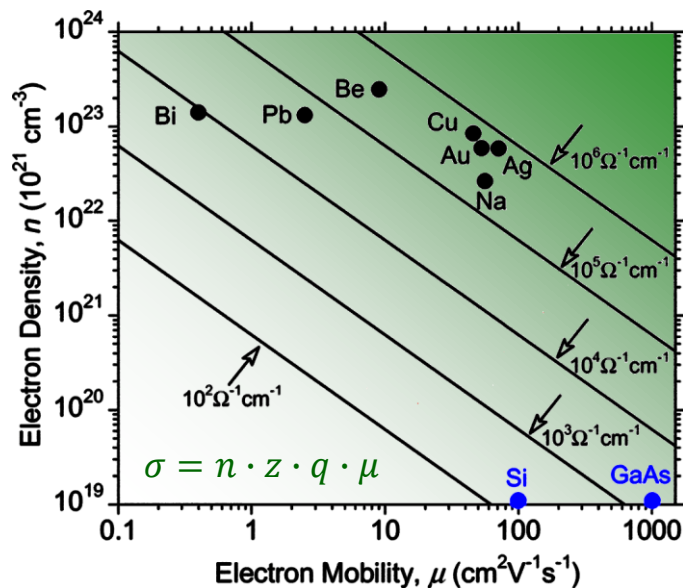
$$\text{with } \frac{v}{E} = \mu$$

$\mu$  – charge carrier mobility /  $cm^2 \cdot V^{-1} \cdot s^{-1}$

results for the electrical conductivity:

$$\sigma = n \cdot z \cdot q \cdot \mu$$

since  $q$  = constant, the electrical conductivity depends on the number of charge carriers  $n$  and their mobility in the material  $\mu$



**(electric) conductivity of selected materials, in  $\text{S m}^{-1}$**

*metals (typical)*

|    |                   |
|----|-------------------|
| Na | $21,3 \cdot 10^6$ |
| Mg | $22,5 \cdot 10^6$ |
| Al | $37,7 \cdot 10^6$ |
| Fe | $10,0 \cdot 10^6$ |
| Cu | $59,8 \cdot 10^6$ |
| Ag | $68,0 \cdot 10^6$ |
| Au | $42,6 \cdot 10^6$ |

*4. main group*

|             |                                   |
|-------------|-----------------------------------|
| C (diamond) | $< 1,0 \cdot 10^{-16}$            |
| Si          | $5,0 \cdot 10^{-4}$               |
| Ge          | $2,0 \cdot 10^0$                  |
| Sn          | $8,7 \cdot 10^6$                  |
| Pb          | $4,7 \cdot 10^6$                  |
| C (grafite) | $3,0 \cdot 10^2 - 3,0 \cdot 10^6$ |

*plastics*

|              |                           |
|--------------|---------------------------|
| polyethylene | $10^{-13}$                |
| polystyrole  | $10^{-17} \dots 10^{-13}$ |

*ceramics*

|                                       |                                                              |
|---------------------------------------|--------------------------------------------------------------|
| $\text{Al}_2\text{O}_3$               | $10^{-12}$                                                   |
| BN                                    | $10^{-11}$                                                   |
| $\text{RuO}_2$                        | $2,5 \cdot 10^6$                                             |
| $\beta\text{-NaAl}_{11}\text{O}_{17}$ | $1,4 \cdot 10^0$ – $\text{Na}^+$ ion conductor, RT           |
| $\text{ZrO}_2\text{:}8\text{Y}$       | $1,5 \cdot 10^1$ – (1000 °C) – $\text{O}^{2-}$ ion conductor |

|                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>conductors:</i> $\sigma > 1 \cdot 10^6 \text{ S m}^{-1}$<br><i>semiconductors:</i> $1 \cdot 10^{-10} \text{ S m}^{-1} < \sigma < 1 \cdot 10^6 \text{ S m}^{-1}$<br><i>insulators:</i> $\sigma < 1 \cdot 10^{-10} \text{ S m}^{-1}$ |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Ion conductor**

conductivity through transport of charged species ( $\rightarrow$  ions)

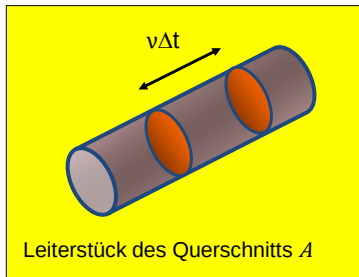
cation conductor: small cations move through sublattices of large anions;

example: AgI

anion conductor: anions move through vacancy hopping;

example: YSZ

## ion conductivity – a microscopic view



$$\text{current } I = \frac{\text{charge transport through area } A}{\text{time intervall } \Delta t}$$

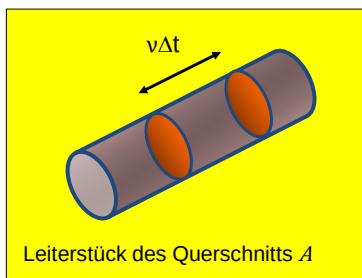
$$\Rightarrow I = \frac{\overbrace{A \cdot v \cdot \Delta t}^{\text{volume}} \cdot \overbrace{z \cdot q \cdot n}^{\text{charge}}}{\Delta t}$$

$z$  – charge number (ions can have multiple charges!)

$q$  – elementary charge

$n$  – number of charge carriers moving with the speed  $v$  in the conductor

## ion conductivity – a microscopic view



$$\text{current density: } \vec{J} = \frac{I}{A} = n \cdot z \cdot q \cdot v$$

conductivity  
(Ohm's law):

$$\sigma = \vec{J} / \vec{E}$$

the following applies  
to the average speed  
( $\mu$  – mobility of the  
charge carriers):

$$v = \mu \cdot \vec{E}$$



valid for ionic and electronic conductors  
→ charge transport by charged species with  
charge  $z \cdot q$  (ions) or electrons ( $z = 1$ )

$$\sigma = n \cdot z \cdot q \cdot \mu$$

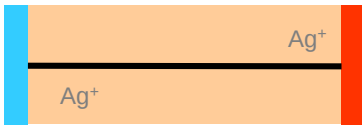




## ion conductivity – a microscopic view

– potential and concentration gradient

driving forces; example: mobility of  $\text{Ag}^+$  in AgI in the electric field  $E$  and in the concentration gradient  $dn/dx$



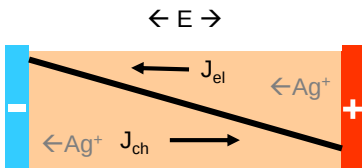
$$J_{el} = n \cdot \mu \cdot \frac{d\varphi}{dx} = 0 \quad \text{potential gradient}$$

$$J_{ch} = -D_T \cdot \frac{dn}{dx} = 0 \quad \text{concentration gradient, } D_T = \text{diffusion coefficient}$$

## ion conductivity – a microscopic view

– potential and concentration gradient

driving forces; example: mobility of  $\text{Ag}^+$  in AgI in the electric field  $E$  and in the concentration gradient  $dn/dx$



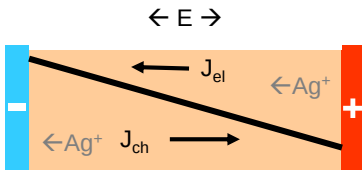
$$J_{el} = n \cdot \mu \cdot \frac{d\varphi}{dx} \neq 0 \quad \text{potential gradient}$$

$$J_{ch} = -D_T \cdot \frac{dn}{dx} \neq 0 \quad \text{concentration gradient, } D_T = \text{diffusion coefficient}$$

## ion conductivity – a microscopic view

– potential and concentration gradient

driving forces; example: mobility of  $\text{Ag}^+$  in AgI in the electric field  $E$  and in the concentration gradient  $dn/dx$



mobility of  $\text{Ag}^+$

$$J_{el} = n \cdot \mu \cdot \frac{d\phi}{dx} \neq 0 \quad \text{potential gradient}$$

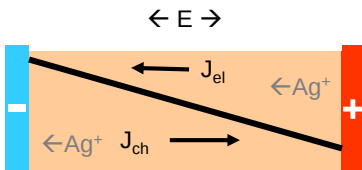
$$J_{ch} = -D_T \cdot \frac{dn}{dx} \neq 0 \quad \text{concentration gradient, } D_T = \text{diffusion coefficient}$$

driving force

## ion conductivity – a microscopic view

– potential and concentration gradient

driving forces; example: mobility of  $\text{Ag}^+$  in AgI in the electric field  $E$  and in the concentration gradient  $dn/dx$



mobility of  $\text{Ag}^+$

$$J_{el} = n \cdot \mu \cdot \frac{d\phi}{dx} \neq 0 \quad \text{potential gradient}$$

$$J_{ch} = -D_T \cdot \frac{dn}{dx} \neq 0 \quad \text{concentration gradient, } D_T = \text{diffusion coefficient}$$

driving force

electrochemical potential for  $\text{Ag}^+$

$$\tilde{\mu}_{\text{Ag}^+} = \mu_{\text{Ag}^+} + z \cdot q \cdot \phi$$

chemical potential  $\text{Ag}^+$

electrir potential

$$\frac{d\tilde{\mu}_{\text{Ag}^+}}{dx} = \frac{d\mu_{\text{Ag}^+}}{dx} + z \cdot q \cdot \frac{d\phi}{dx} = 0 \Rightarrow \frac{d\mu_{\text{Ag}^+}}{dx} = -z \cdot q \cdot \frac{d\phi}{dx}$$

in equilibrium:

$$J_{el} = J_{ch}$$

the gradient of the electrochemical potential in the equilibrium is zero

## ion conductivity – a microscopic view

– potential and concentration gradient

driving forces; example: mobility of  $\text{Ag}^+$  in AgI in the electric field E and in the concentration gradient  $dn/dx$

$$J_{el} = n \cdot \mu \cdot \frac{d\phi}{dx}$$

$$J_{ch} = -D_T \cdot \frac{dn}{dx}$$

generally applies:  
the chemical potential is  
concentration-dependent..

concentration gradient is  
converted into a chemical  
potential gradient

$$\mu_{\text{Ag}^+} = k \cdot T \cdot \ln n$$

$$\frac{d\mu_{\text{Ag}^+}}{dx} = k \cdot T \cdot \frac{1}{n} \frac{dn}{dx}$$

$k$  = Boltzmann constant,  $1,3806 \cdot 10^{-23} \text{ J K}^{-1}$

## ion conductivity – a microscopic view

– potential and concentration gradient

driving forces; example: mobility of  $\text{Ag}^+$  in AgI in the electric field E and in the concentration gradient  $dn/dx$

$$J_{el} = n \cdot \mu \cdot \frac{d\phi}{dx}$$

$$J_{ch} = -D_T \cdot \frac{dn}{dx}$$

generally applies:  
the chemical potential is  
concentration-dependent..

concentration gradient is  
converted into a chemical  
potential gradient

$$\mu_{\text{Ag}^+} = k \cdot T \cdot \ln n$$

$$\frac{d\mu_{\text{Ag}^+}}{dx} = k \cdot T \cdot \frac{1}{n} \frac{dn}{dx}$$

$$J_{ch} = -D_T \cdot \frac{n}{k \cdot T} \cdot \frac{d\mu_{\text{Ag}^+}}{dx}$$

$k$  = Boltzmann constant,  $1,3806 \cdot 10^{-23} \text{ J K}^{-1}$

## ion conductivity – a microscopic view

– potential and concentration gradient

driving forces; example: mobility of  $\text{Ag}^+$  in  $\text{AgI}$  in the electric field  $E$  and in the concentration gradient  $dn/dx$

$$J_{el} = n \cdot \mu \cdot \frac{d\phi}{dx}$$

$$J_{ch} = -D_T \cdot \frac{dn}{dx}$$

in equilibrium:

$$J_{el} = J_{ch}$$

generally applies:  
the chemical potential is  
concentration-dependent..

concentration gradient is  
converted into a chemical  
potential gradient

$$\mu_{\text{Ag}^+} = k \cdot T \cdot \ln n$$

$$\frac{d\mu_{\text{Ag}^+}}{dx} = k \cdot T \cdot \frac{1}{n} \frac{dn}{dx}$$

$$J_{ch} = -D_T \cdot \frac{n}{k \cdot T} \cdot \frac{d\mu_{\text{Ag}^+}}{dx}$$

$k$  = Boltzmann constant,  $1,3806 \cdot 10^{-23} \text{ J K}^{-1}$

## ion conductivity – a microscopic view

– potential and concentration gradient

since in the equ.:

$$J_{el} = J_{ch}$$

$$\overbrace{-D_T \cdot \frac{n}{k \cdot T} \cdot \frac{d\mu_{\text{Ag}^+}}{dx}}^{\text{chem. Current density } J_{ch}} = \overbrace{n \cdot \mu \cdot \frac{d\phi}{dx}}^{\text{electr. current density } J_{el}}$$

## ion conductivity – a microscopic view

– potential and concentration gradient

chem. Current density  $J_{ch}$     electr. current density  $J_{el}$

since in the equ.:  $J_{el} = J_{ch}$

$$-D_T \cdot \frac{n}{k \cdot T} \cdot \frac{d\mu_{Ag^+}}{dx} = n \cdot \mu \cdot \frac{d\phi}{dx}$$

$$\frac{d\mu_{Ag^+}}{dx} = -z \cdot q \cdot \frac{d\phi}{dx}$$

$$\mu = \frac{z \cdot q}{k \cdot T} \cdot D_T$$

the ion mobility  $\mu$  is linked to the diffusion coefficient  $D_T$ , Nernst-Einstein relation  $D_T = (\mu \cdot k \cdot T)/q$ .  
an increase in temperature leads to a decrease in mobility

## ion conductivity – a microscopic view

– potential and concentration gradient

chem. Current density  $J_{ch}$     electr. current density  $J_{el}$

since in the equ.:  $J_{el} = J_{ch}$

$$-D_T \cdot \frac{n}{k \cdot T} \cdot \frac{d\mu_{Ag^+}}{dx} = n \cdot \mu \cdot \frac{d\phi}{dx}$$

$$\frac{d\mu_{Ag^+}}{dx} = -z \cdot q \cdot \frac{d\phi}{dx}$$

$$\mu = \frac{z \cdot q}{k \cdot T} \cdot D_T$$

$$\sigma = n \cdot z \cdot q \cdot \mu$$

$$\sigma = n \cdot \frac{1}{k \cdot T} \cdot D_T$$

the ion mobility  $\mu$  is linked to the diffusion coefficient  $D_T$ , Nernst-Einstein relation  $D_T = (\mu \cdot k \cdot T)/q$ .  
an increase in temperature leads to a decrease in mobility. An increase in temperature leads to a decrease in mobility and thus conductivity due to an increase in the collision probability of the charge carriers

### ion conductivity – a microscopic view

– potential and concentration gradient

$$\sigma = n \cdot \frac{1}{k \cdot T} \cdot D_T$$

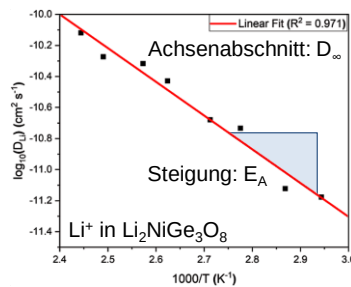
Diffusion in solids is slow and extremely temperature-dependent. The following applies approximately for DT:

$$D_T = D_\infty \cdot \exp\left(\frac{E_A}{RT}\right)$$

$D_\infty$  – diffusion coefficient at  $\rightarrow \infty$   
(intercept in the Arrhenius plot)

$E_A$  – activation energy  
(slope in the arrhenius plot)

$R$  – general gas constant; 8,314 J K<sup>-1</sup> mol<sup>-1</sup>



### ion conductivity – a microscopic view

– potential and concentration gradient

$$\sigma = n \cdot \frac{1}{k \cdot T} \cdot D_T$$



$$\sigma = n \cdot \underbrace{\frac{1}{k \cdot T}}_{\text{Charge carrier mobility}} \cdot \underbrace{D_\infty \cdot \exp\left(\frac{E_A}{R \cdot T}\right)}_{\text{diffusion}}$$

T influence on diffusion (exponential term) outweighs the negative effect on mobility → Ion conductivity increases with temperature increase

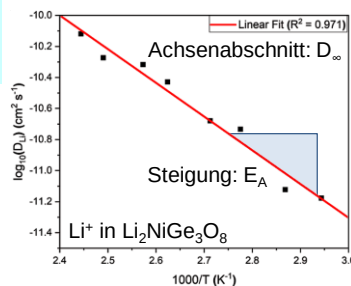
Diffusion in solids is slow and extremely temperature-dependent. The following applies approximately for DT:

$$D_T = D_\infty \cdot \exp\left(\frac{E_A}{RT}\right)$$

$D_\infty$  – diffusion coefficient at  $\rightarrow \infty$   
(intercept in the Arrhenius plot)

$E_A$  – activation energy  
(slope in the arrhenius plot)

$R$  – general gas constant; 8,314 J K<sup>-1</sup> mol<sup>-1</sup>



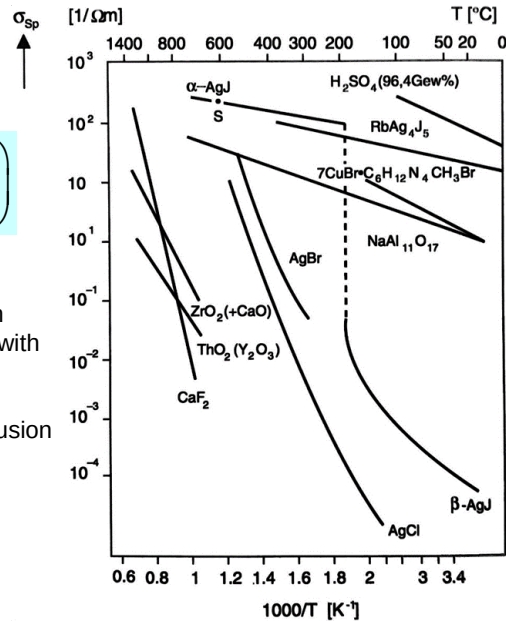
## ion conductivity – a microscopic view

– temperature influence

$$\sigma = n \cdot \frac{1}{k \cdot T} \cdot D_{\infty} \cdot \exp\left(\frac{E_A}{R \cdot T}\right)$$

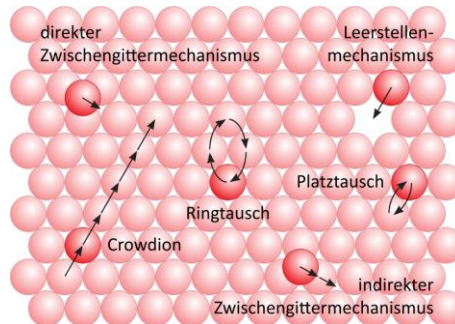
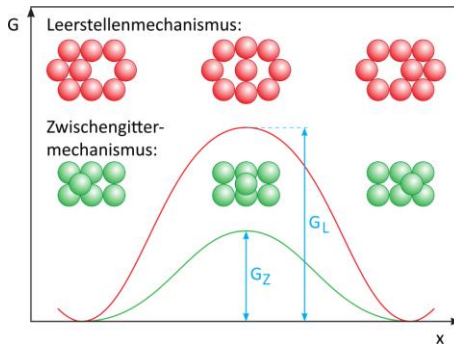
T-Influence on diffusion (exponential term) outweighs the negative effect on mobility → Ion conductivity increases with temperature increase;

in addition to the temperature, the diffusion coefficient also depends on defects present in the solid



## ion conductivity – a microscopic view

– role of defects

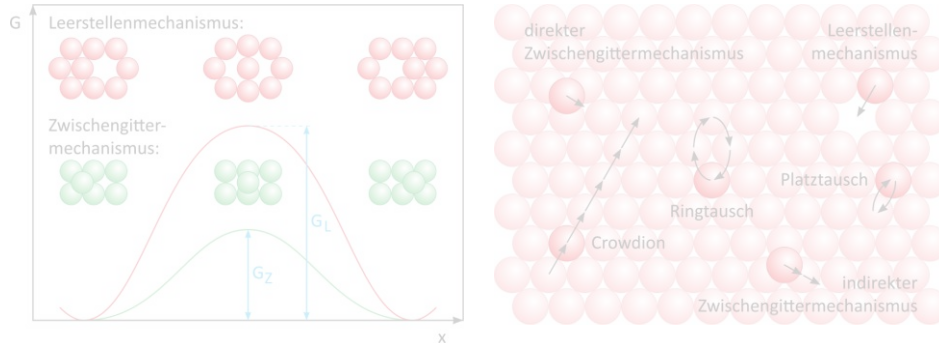


2 basic possibilities for diffusion of ions: vacancy and interstitial mechanism; the activation energy is different and depends largely on the size of the diffusing species

interstitial mechanism  
in bcc Fe:

| diff. spezie | $D_{\infty} / \text{mm}^2 \text{s}^{-1}$ | $E_A / \text{kJ mol}^{-1}$ |
|--------------|------------------------------------------|----------------------------|
| C            | 2,0                                      | 84,1                       |
| N            | 0,3                                      | 76,1                       |
| H            | 0,1                                      | 13,4                       |

## ion conductivity – a microscopic view – role of defects



2 basic possibilities for diffusion of ions: vacancy and interstitial mechanism; the activation energy is different and depends largely on the size of the diffusing species

interstitial mechanism  
in bcc Fe:

| diff. spezie | $D_{\infty} / \text{mm}^2 \text{s}^{-1}$ | $E_A / \text{kJ mol}^{-1}$ |
|--------------|------------------------------------------|----------------------------|
| C            | 2,0                                      | 84,1                       |
| N            | 0,3                                      | 76,1                       |
| H            | 0,1                                      | 13,4                       |

## Ionenleitfähigkeit mikroskopisch – Rolle von Defekten



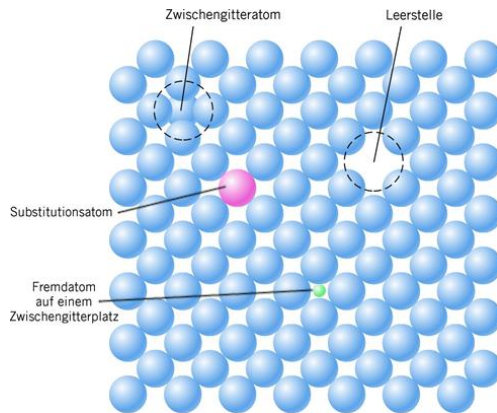
2 prinzipielle Möglichkeiten der Diffusion von Ionen: Leerstellen- und Zwischengittermechanismus; die Aktivierungsenergie ist unterschiedlich und hängt maßgeblich von der Größe der diffundierenden Spezies ab

Zwischengittermechanismus  
in krz Fe:

| diff. Spezies | $D_{\infty} / \text{mm}^2 \text{s}^{-1}$ | $E_A / \text{kJ mol}^{-1}$ |
|---------------|------------------------------------------|----------------------------|
| C             | 2,0                                      | 84,1                       |
| N             | 0,3                                      | 76,1                       |
| H             | 0,1                                      | 13,4                       |



**repetition:** defects and imperfections -> zero-dimensional point defects



*vacancy:*  
atom on regular lattice site is missing!

*interstitial atom:*  
atom on interstitial site

*substitutional atom:*  
doping on regular lattice site  
doping on interstitial lattice site

in ionic crystals, defects can occur equally in cation and anion sublattices.  
Electroneutrality must be maintained!

description of defects via Kroeger-Vink notation: reaction equations with defects

$$M_S^C$$

*M:* species (which atom/ion is considered), *v* = vacancy, *e* = electron

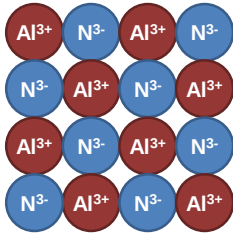
*S:* lattice site on which *M* sits; *i* = interstitial

*C:* charge relative to regular species on lattice site =  $C_M - C_S$

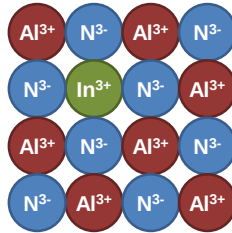
x = neutral  
• = 1x positiv  
• • = 2x positiv  
/ = 1x negativ  
// = 2x negativ

Kröger-Vink Notation: example AlN

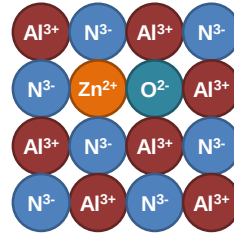
$\text{Zn}^{2+}$  has a + charge less than  $\text{Al}^{3+} \rightarrow$  deficit in negative charge



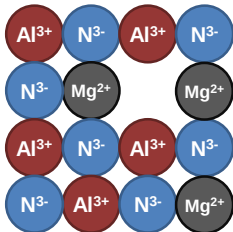
$\text{Al}_{\text{Al}}^x$  (trivial)



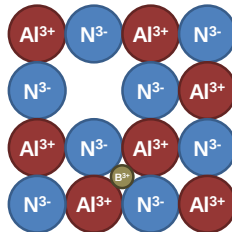
$\text{In}_{\text{Al}}^x$



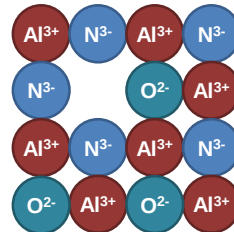
$\text{Zn}_{\text{Al}}' + \text{O}_{\text{N}}^\bullet$



$3\text{Mg}_{\text{Al}}' + v_{\text{N}}^{\bullet\bullet}$

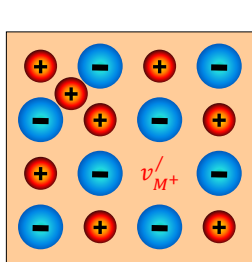


$v_{\text{Al}}^{///} + \text{B}_i^{\bullet\bullet\bullet}$

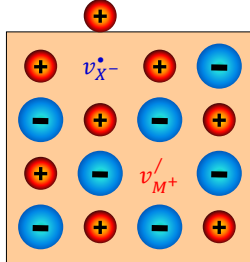


$v_{\text{Al}}^{///} + 3\text{O}_{\text{N}}^\bullet$

**defect chemistry:** often very specific combinations of different defects in the anion and cation sublattice: intrinsic defects



a) Frenkel defect

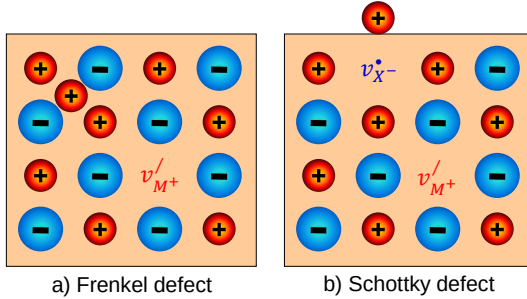


b) Schottky defect

a) *Frenkel defect:*  $v_{\text{M}^+}' + \text{M}_i^*$   
more likely for small ions  $\rightarrow$   
cation sublattice  
example: Ag halides

b) *Schottky defect:*  $v_{\text{M}^+}' + v_{\text{X}^-}^*$   
in compounds without  
favorable interstitial sites  
example: alkali halides

**defect chemistry:** often very specific combinations of different defects in the anion and cation sublattice: intrinsic defects



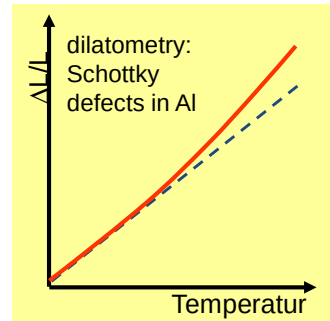
- a) *Frenkel defect:*  $v'_{M^+} + M_i^*$   
more likely for small ions  $\rightarrow$   
cation sublattice  
example: Ag halides
- b) *Schottky defect:*  $v'_{M^+} + v_{X^-}^*$   
in compounds without  
favorable interstitial sites  
example: alkali halides

no significant increase  
in volume, more likely  
for very different ionic  
radii  $r_{\text{Anion}}/r_{\text{Kation}}$

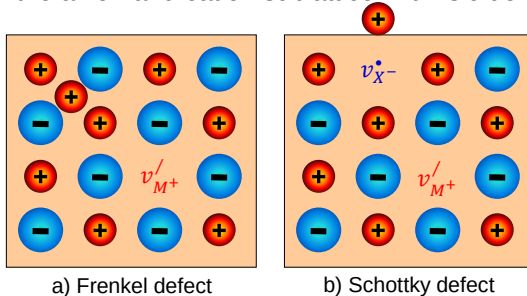
ions accumulate on the  
crystal surface, the  
volume increases,  
preferably for similar  
ion radii  $r_{\text{Anion}} \approx r_{\text{Kation}}$

$\text{Ag}^+ = 129 \text{ pm}$   
 $\text{Br}^- = 182 \text{ pm}$   
 $\text{I}^- = 206 \text{ pm}$

$\text{K}^+ = 152 \text{ pm}$   
 $\text{Cl}^- = 167 \text{ pm}$   
 $\text{Br}^- = 182 \text{ pm}$



**defect chemistry:** often very specific combinations of different defects in the anion and cation sublattice: intrinsic defects



- a) *Frenkel defect:*  $v'_{M^+} + M_i^*$   
more likely for small ions  $\rightarrow$   
cation sublattice  
example: Ag halides
- b) *Schottky defect:*  $v'_{M^+} + v_{X^-}^*$   
in compounds without  
favorable interstitial sites  
example: alkali halides

For the formation of intrinsic defects, the MWG can be formulated with the equilibrium constant  $K = c(\text{products})/c(\text{reactants})$ : *Frenkel:*  $\text{Ag}_{\text{Ag}}^x \rightleftharpoons v'_{\text{Ag}} + \text{Ag}_i^*$

*Schottky:*  $K_K^x + \text{Cl}_{\text{Cl}}^x + v'_{\text{K,OF}} + v_{\text{Cl,OF}}^* \rightleftharpoons v'_{\text{K}} + v_{\text{Cl}}^* + K_{\text{K,OF}}^x + \text{Cl}_{\text{Cl,OF}}^x$

Mit  $\Delta G^0 = -RT \cdot \ln K$  results for the defect concentration  $c_F$  resp.  $c_S$ :

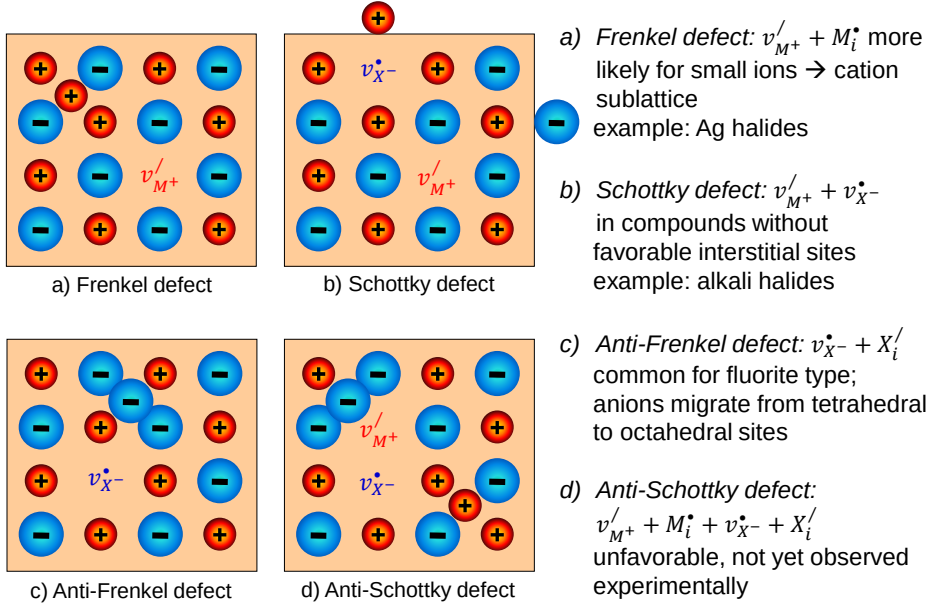
*Frenkel:*  $c_F = \sqrt{N/N'} \cdot \exp\left(-\frac{E_A}{2kT}\right)$

*Schottky:*  $c_S = N \cdot \exp\left(-\frac{E_A}{2kT}\right)$

► defect concentration is  $f(T)$

$N/N'$  = number regular/ interstitial position

**defect chemistry:** often very specific combinations of different defects in the anion and cation sublattice: intrinsic defects



**defect chemistry:** defectconcentration as a function of temperature

| Defektkonzentration | 1 eV              | 2 eV               | 4 eV               | 6 eV               | 8 eV               |
|---------------------|-------------------|--------------------|--------------------|--------------------|--------------------|
| $n/N$ bei 100°C     | $2 \cdot 10^{-7}$ | $3 \cdot 10^{-14}$ | $1 \cdot 10^{-27}$ | $3 \cdot 10^{-41}$ | $1 \cdot 10^{-54}$ |
| $n/N$ bei 500°C     | $6 \cdot 10^{-4}$ | $3 \cdot 10^{-7}$  | $1 \cdot 10^{-13}$ | $3 \cdot 10^{-20}$ | $8 \cdot 10^{-27}$ |
| $n/N$ bei 800°C     | $4 \cdot 10^{-3}$ | $2 \cdot 10^{-5}$  | $4 \cdot 10^{-10}$ | $8 \cdot 10^{-15}$ | $2 \cdot 10^{-19}$ |
| $n/N$ bei 1000°C    | $1 \cdot 10^{-2}$ | $1 \cdot 10^{-4}$  | $1 \cdot 10^{-8}$  | $1 \cdot 10^{-12}$ | $1 \cdot 10^{-16}$ |
| $n/N$ bei 1200°C    | $2 \cdot 10^{-2}$ | $4 \cdot 10^{-4}$  | $1 \cdot 10^{-7}$  | $5 \cdot 10^{-11}$ | $2 \cdot 10^{-14}$ |
| $n/N$ bei 1500°C    | $4 \cdot 10^{-2}$ | $1 \cdot 10^{-4}$  | $2 \cdot 10^{-6}$  | $3 \cdot 10^{-9}$  | $4 \cdot 10^{-12}$ |
| $n/N$ bei 1800°C    | $6 \cdot 10^{-2}$ | $4 \cdot 10^{-3}$  | $1 \cdot 10^{-5}$  | $5 \cdot 10^{-8}$  | $2 \cdot 10^{-10}$ |
| $n/N$ bei 2000°C    | $8 \cdot 10^{-2}$ | $6 \cdot 10^{-3}$  | $4 \cdot 10^{-5}$  | $2 \cdot 10^{-7}$  | $1 \cdot 10^{-9}$  |

AgBr: 1,1 eV  
(Frenkel)

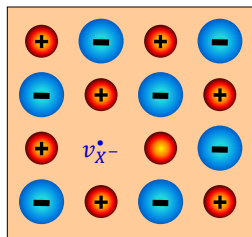
NaCl: 2,3 eV  
(Schottky)

CaO: 6 eV  
(Schottky)

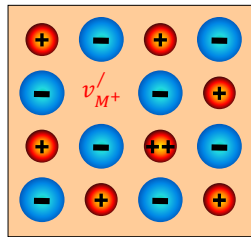
CaF<sub>2</sub>: 2,5 eV  
(Anti-Frenkel)

1,0 g:  $N \approx 1 \cdot 10^{21} - 1 \cdot 10^{22}$

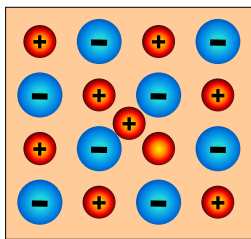
**Defect chemistry:** extrinsic defects are created by doping with foreign material (possibly a different charge):



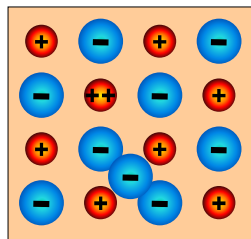
a) anion deficit



b) cation deficit



c) cation excess



d) anion excess

a) Anion deficit:  $v_{X-}^{\bullet} + M_{M+}^{0/}$   
example: ZnO, YSZ ( $Zr^{4+} \rightarrow Y^{3+}$ )

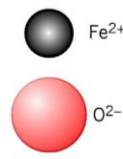
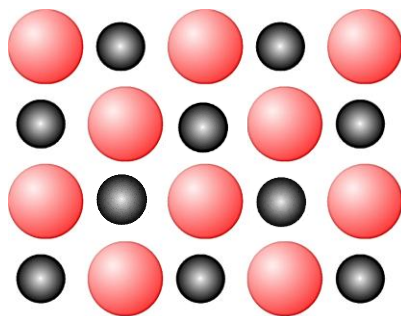
b) Cation deficit:  $v_{M+}^{\prime} + M_{M+}^{2+\bullet}$   
example: ZnO:Al<sup>3+</sup>

c) cation excess:  
 $M_i^{\bullet} + M_{M+}^{0/}$

d) anion excess:  
 $M_{M+}^{2+\bullet} + X_i^{\prime}$   
example: In<sub>2</sub>O<sub>3</sub>:Sn<sup>4+</sup>

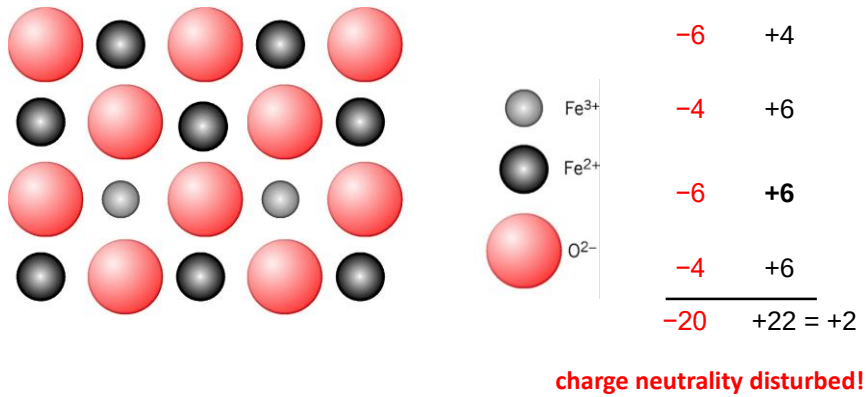
► defect concentration is temperature-independent and only determined by the dopant concentration

**defect chemistry:** example: FeO; phase width Fe<sub>1-δ</sub>O by partial oxidation of Fe<sup>2+</sup> to Fe<sup>3+</sup>

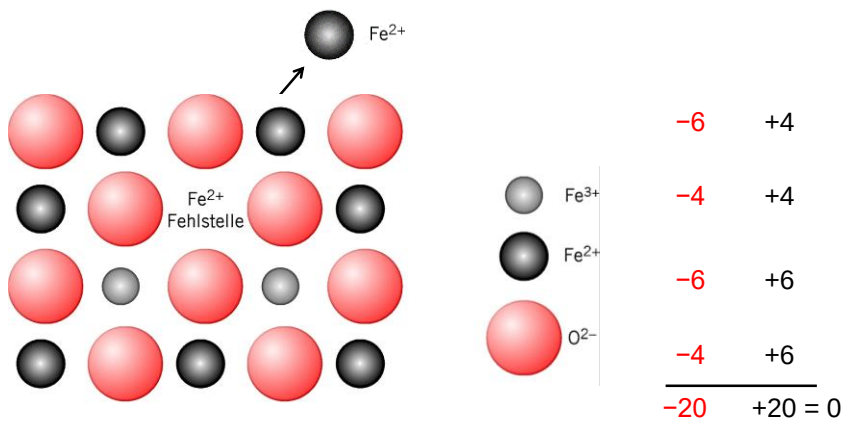


|       |         |
|-------|---------|
| -6    | +4      |
| -4    | +6      |
| -6    | +4      |
| -4    | +6      |
| <hr/> |         |
| -20   | +20 = 0 |

**defect chemistry: example:** FeO; phase width  $\text{Fe}_{1-\delta}\text{O}$  by partial oxidation of  $\text{Fe}^{2+}$  to  $\text{Fe}^{3+}$

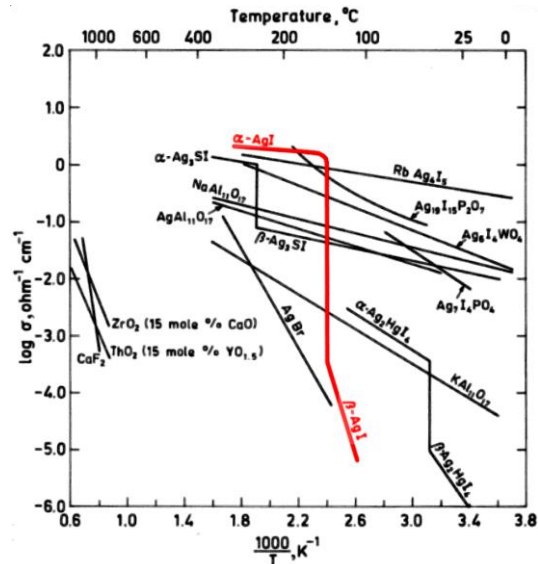


**defect chemistry: example:** FeO; phase width  $\text{Fe}_{1-\delta}\text{O}$  by partial oxidation of  $\text{Fe}^{2+}$  to  $\text{Fe}^{3+}$



**ion conductors:** silver halides are among the best ionic conductors, especially at low temperatures

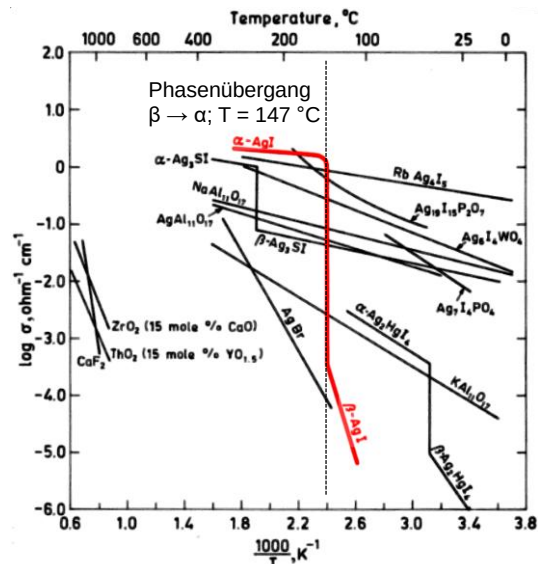
→ fast ionic conductors, superionic conductors

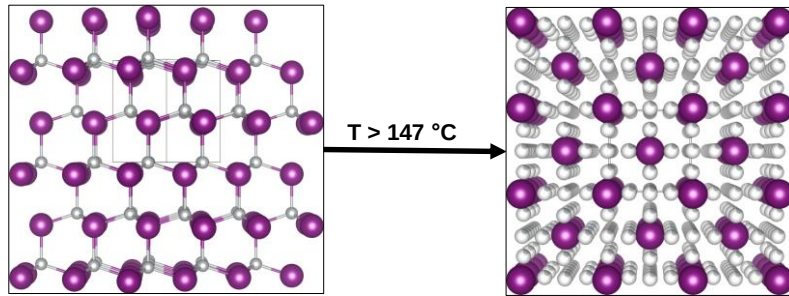


**ion conductors:** silver halides are among the best ionic conductors, especially at low temperatures

→ fast ionic conductors, superionic conductors

**silver(I)-iodide:**  $\beta\text{-AgI}$  stable up to  $147^{\circ}\text{C}$  (wurtzite), above  $\alpha\text{-AgI}$  (cubic) with disordered Ag layers → “molten” cation sublattice → conductivity increases by 4 orders of magnitude; can be stabilized by doping  $\alpha$ -phase





examples:

$\beta$ -alumina as a membrane for NaS batteries  
SOFC membrane as a oxygen ion conductor



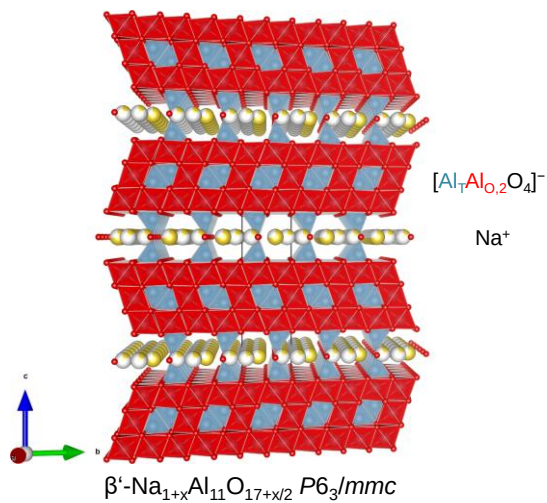
### example-1:

#### $\beta$ -alumina

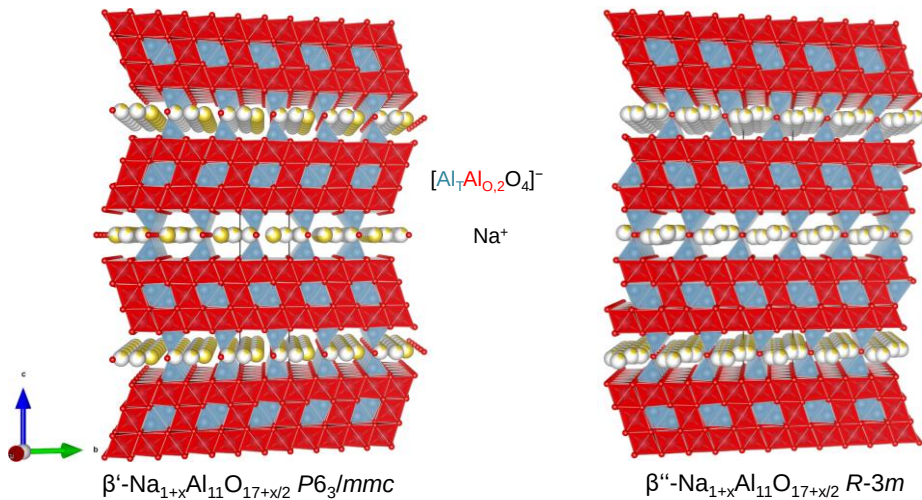
**Ion conductor:**  $\beta\text{-Na}_{1+x}\text{Al}_{11}\text{O}_{17+x/2}$  ( $x \approx 0,22$ )

structure consists of spinel blocks  $[\text{Al}_7\text{Al}_{0,2}\text{O}_4]^-$  with embedded  $\text{Na}^+$  ions

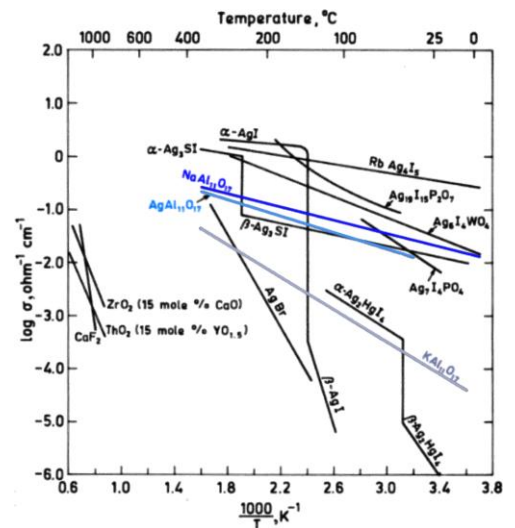
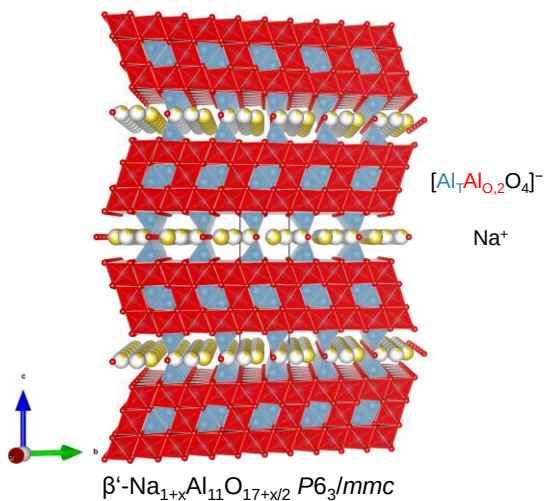
$\text{Na}^+$  in ab plane very mobile  $\rightarrow \text{Na}^+$  ion conductor ( $\approx 30 \text{ S m}^{-1}$  at  $300^\circ\text{C}$ ), band gap of 9 eV  $\rightarrow$  electronic insulator



Na<sup>+</sup> in ab plane very mobile → Na<sup>+</sup> ion conductor ( $\approx 30 \text{ S m}^{-1}$  at 300 °C), band gap of 9 eV → electronic insulator

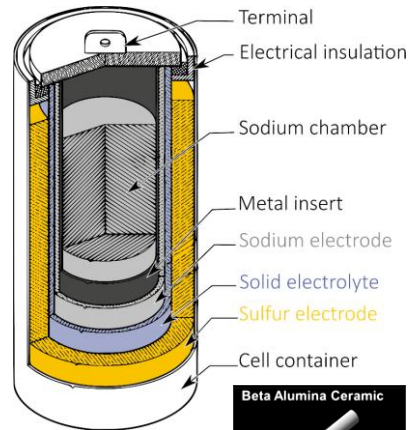


Na<sup>+</sup> in ab plane very mobile → Na<sup>+</sup> ion conductor ( $\approx 30 \text{ S m}^{-1}$  at 300 °C), band gap of 9 eV → electronic insulator



**ion conductor:**  $\beta\text{-Na}_{1+x}\text{Al}_{11}\text{O}_{17+x/2}$  ( $x \approx 0,22$ )

application: separator in high-temperature Na-S batteries

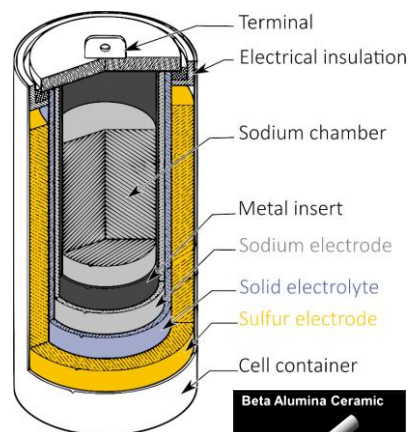
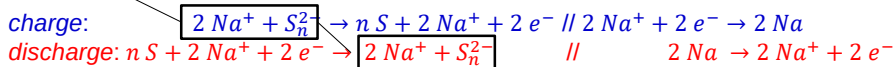
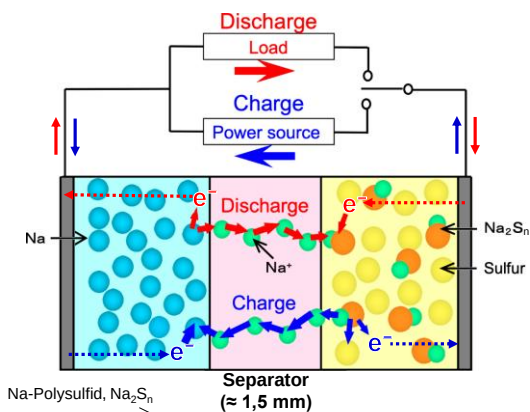


50 x 9 x 0,15 cm

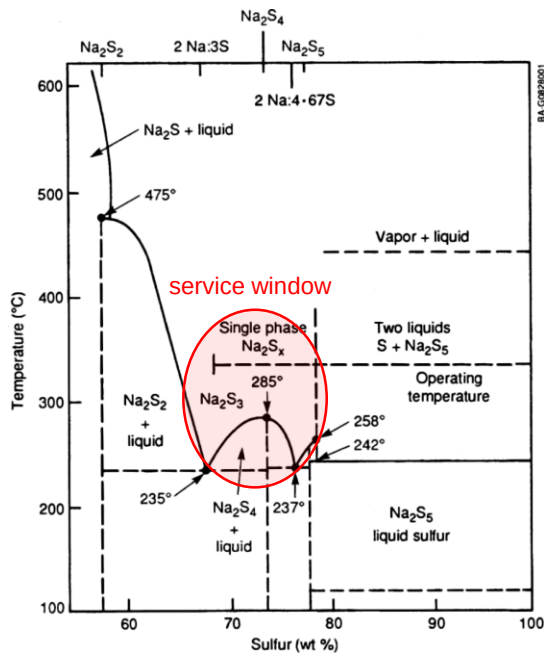
**ion conductor:**  $\beta\text{-Na}_{1+x}\text{Al}_{11}\text{O}_{17+x/2}$  ( $x \approx 0,22$ )

application: separator in high-temperature Na-S batteries

voltage:  $\approx 2,08$  V; energy density:  $> 200$  Wh  $\text{kg}^{-1}$  (in comparison: Li-Ion, 5x Pb/PbSO<sub>4</sub>)



50 x 9 x 0,15 cm

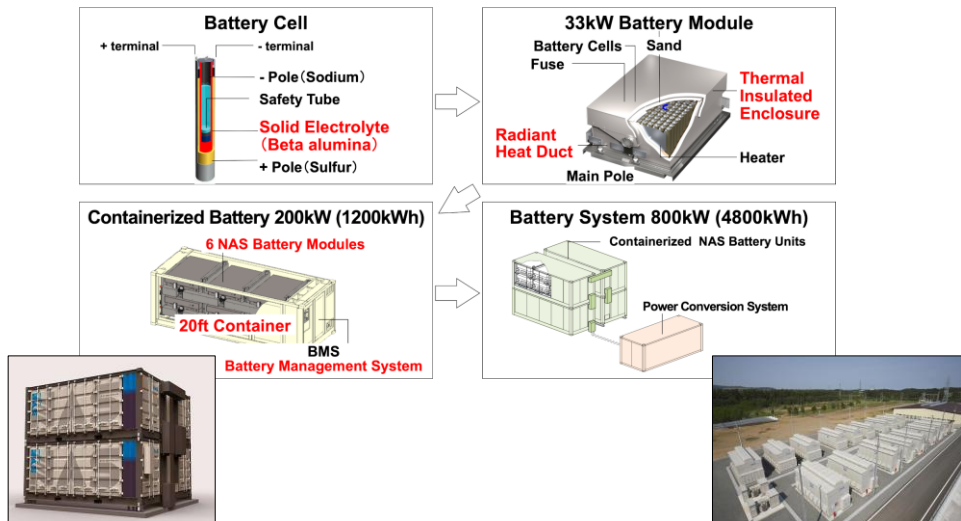


**ion conductor:**  $\beta\text{-Na}_{1+x}\text{Al}_{11}\text{O}_{17+x/2}$  ( $x \approx 0,22$ )

application: separator in high-temper Na-S batteries

voltage:  $\approx 2,08$  V; energy density:  $> 200$  Wh kg<sup>-1</sup> (in comparison: Li-Ion, 5x Pb/PbSO<sub>4</sub>)

→ stationary energy storage



**ion conductor:**  $\beta\text{-Na}_{1+x}\text{Al}_{11}\text{O}_{17+x/2}$  ( $x \approx 0,22$ )

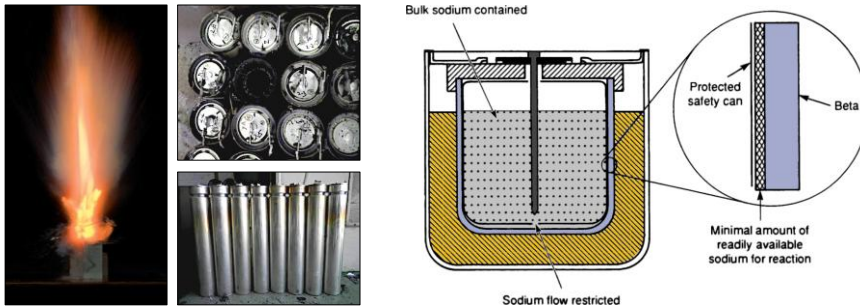
application: separator in high-temptr Na-S batteries

voltage:  $\approx 2,08$  V; energy density:  $> 200$  Wh  $\text{kg}^{-1}$  (in comparison: Li-Ion, 5x Pb/PbSO<sub>4</sub>)

→ stationary energy storage

**safety aspect:** Sodium and sulphur are highly reactive, especially at operating temperature!  
Worst case: Breakage of the  $\beta\text{-NaAl}_{11}\text{O}_{17}$  separator

**avoidance strategies:** Minimize the amount of Na near the separator, encapsulate cells to protect surrounding cells



**example-2:**

YSZ

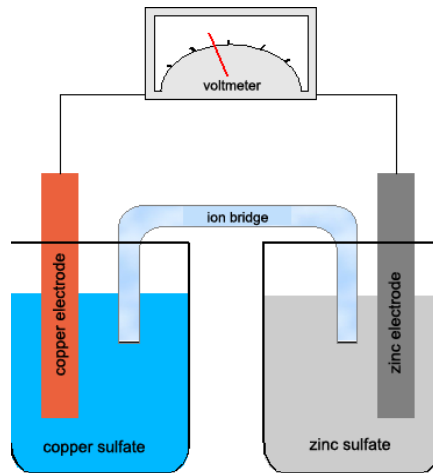
### excursion: galvanic element

A *galvanic cell* or a *galvanic element* is a device for the transformation of chemical energy into electrical energy; application in batteries (primary and secondary)

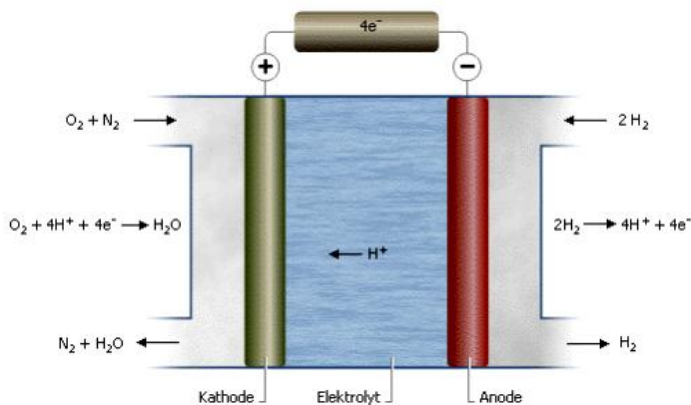
A combination of *two different electrodes* and *an electrolyte* is called a *galvanic element*; it serves as a direct voltage source. The characteristic value is the component voltage/ imposed voltage (Teilspannung/ eingeprägte Spannung).

Capacity of the galvanic element: product of the discharge current and discharge time.

Working principle: redox reaction, at which reduction and oxidation are occurring in spatially separated half cells (half elements).



### Schematic setup and working principle of fuel cells



Anode reaction:  $2 \text{H}_2 \rightarrow 4 \text{H}^+ + 4 \text{e}^-$

Cathode (Kathode) reaction:  $4 \text{H}^+ + \text{O}_2 + 4 \text{e}^- \rightarrow 2 \text{H}_2\text{O}$

**Overall reaction:**  $2 \text{H}_2 + \text{O}_2 \rightarrow 2 \text{H}_2\text{O}$

$\Delta H^0 = -537 \text{ kJ mol}^{-1}$



## Requirements for the components of a fuel cell

### **Electrolyte**

ion transport, separation of the gas volumes, gas-tight, stable in red-ox atmosphere  
(ion conductivity  $> 10 \text{ S/m}$ )

### **Cathode**

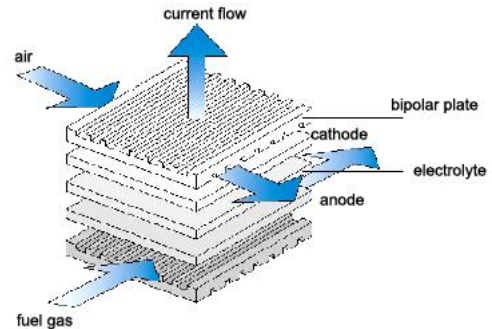
gas transport to electrolyte interface, gas-permeable, electron transport, stable in **oxidizing atmosphere**, electronic or mixed conductivity  $> 1000 \text{ S/m}$

### **Anode**

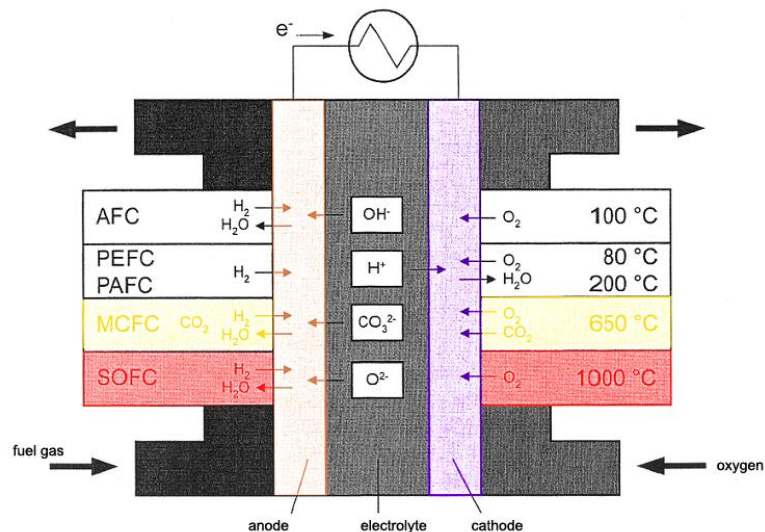
gas transport to electrolyte interface, gas-permeable, electron transport, stable in **reducing atmosphere**, electronic or mixed conductivity  $> 1000 \text{ S/m}$

### **Bipolar plate**

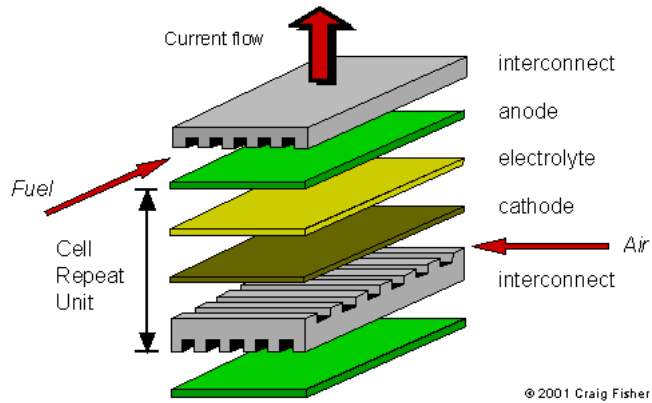
(connects cells to *stacks*), separation of gas volumes, electron transport, electronic conductivity  $> 1000 \text{ S/m}$ , stable in oxidizing and reducing atmosphere



## Principle of the different types of fuel cells

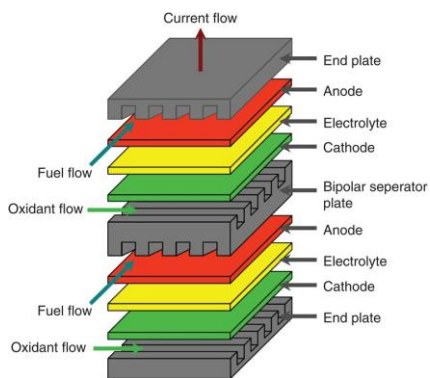


## SOFC

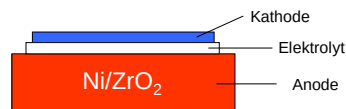


### Aufbau von SOFCs

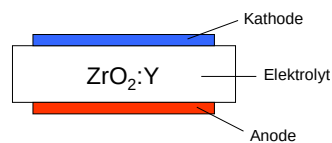
#### Planarstruktur



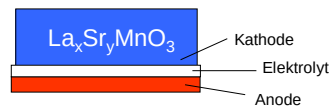
anodengeträgert:  $\text{Ni/ZrO}_2$ -Cermet  
→ mech. Festigkeit



elektrolytgeträgert:  $\text{ZrO}_2\text{:Y}$   
→ mech. Festigkeit

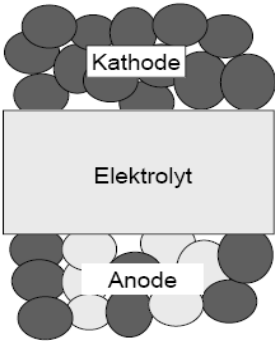


kathodengeträgert:  $\text{La}_x\text{Sr}_y\text{MnO}_3$  (LSM)  
→ mech. Festigkeit

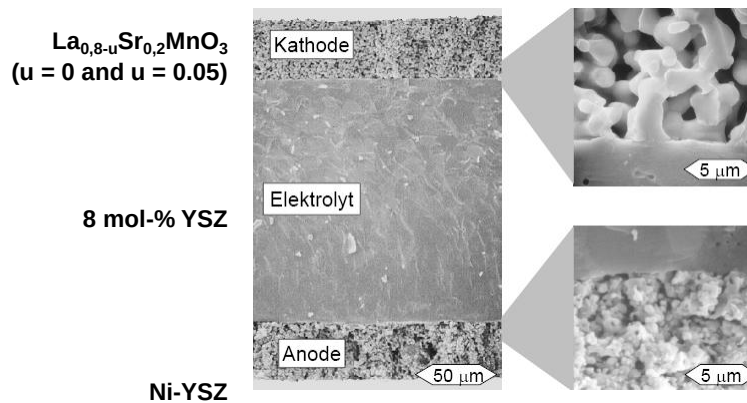




## Function of the single components of a SOFC

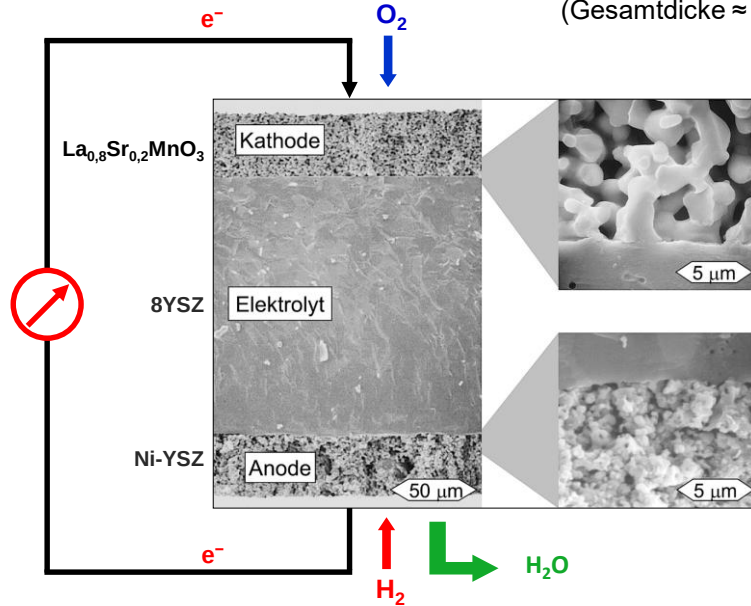
| Material                                                        |                                                                                   | Electrical parameters                                                    | Thermo-chemical parameters                    |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------|
| strontium doped lanthanum manganate<br>(La, Sr)MnO <sub>3</sub> |  | el conductivity<br>$(\sigma_{el} + \sigma_{ion})$                        | porosity<br>CTE                               |
| 8 mol-% YSZ                                                     |                                                                                   | catalytic activity<br>Ion conductivity<br>$\sigma_{ion} \gg \sigma_{el}$ | adhesion<br>gas-tightness<br>Stability        |
| Ni-YSZ                                                          |                                                                                   | catalytic activity<br>el conductivity<br>$(\sigma_{el} + \sigma_{ion})$  | adhesion<br>expansion coefficient<br>porosity |
|                                                                 |                                                                                   |                                                                          |                                               |
|                                                                 |                                                                                   |                                                                          |                                               |

## SEM image of a SOFC solid electrolyte with cathode and anode

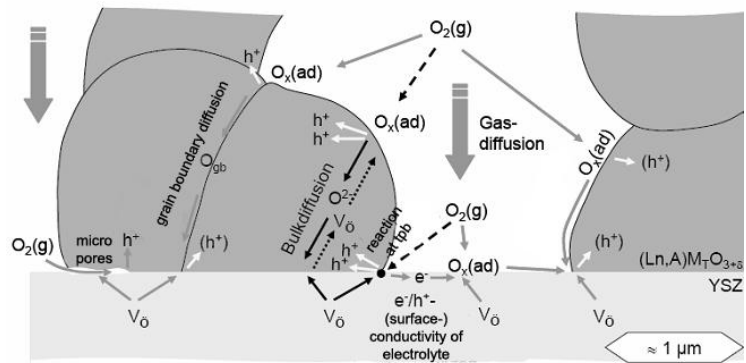


## Aufbau von SOFCs:

REM-Aufnahme eines SOFC Festkörperelektrolyt mit Kathode und Anode  
(Gesamtdicke  $\approx 250 \mu\text{m}$ )



## Reaction pathways of oxygen during reduction at the cathode



## Requirements for cathode

- high electric conductivity
- open porosity
- high electro-catalytic activity for  $\text{O}_2$ -reduction
- chemically stable in oxidizing atmosphere up to  $1000^\circ\text{C}$
- chemically and thermo-mechanically compatible with solid electrode
- long-term stable microstructure (no post sintering at high temperatures allowed)
- structural stability at load reversal

**cathode material:** (La,Sr)MnO<sub>3</sub>

mixed conductive metal oxides, usually perovskites (ABO<sub>3</sub>) of the general composition LnM<sub>T</sub>O<sub>3</sub>

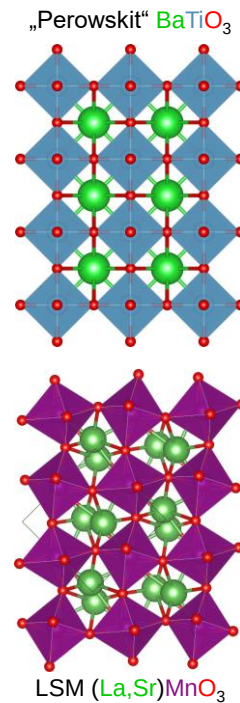
Ln - lanthanide (mainly La) M<sub>T</sub> - transition metal (Cr, Mn, Fe, Co, Ni)

doping with M<sup>2+</sup> (Ca, Sr) on A-layer, charge compensation by oxidation of Mn<sup>3+</sup> → Mn<sup>4+</sup> → conductivity via vacancies

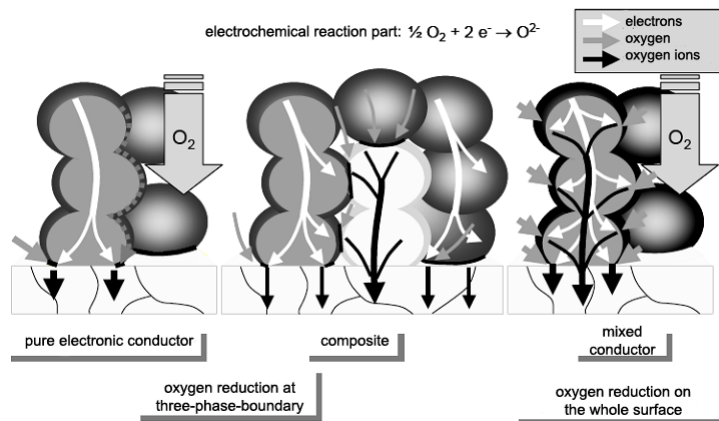
frequently used material: LSM (La,Sr)MnO<sub>3</sub> with 15 - 30 % Sr content

increase in electronic and ionic conductivity and catalytic activity due to Fe, Co, Ni on B-site of Sr- and Ca-doped LSMs

oxygen ion conductivity or diffusion coefficient increases with increasing doping on the A site



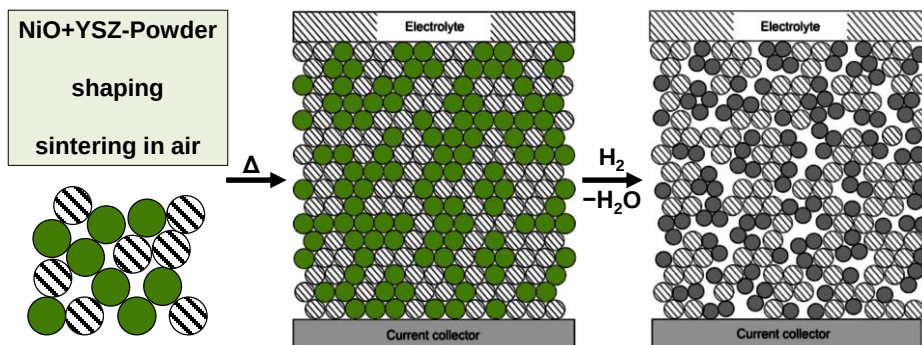
### Types of electronic-conducting and mixed-conducting cathodes



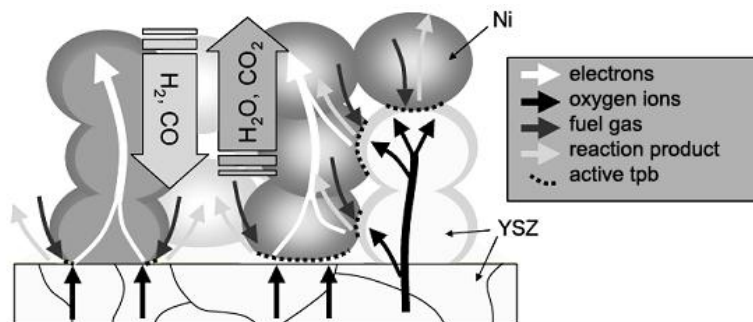
While the standard cathode material LSM has only a neglectable ionic conductivity and the electrochemical reactions are restricted to the three-phase-boundary, mixed-conductive cathode materials enable the extension of the electro-chemical active zone onto the entire surface. In case of a composite-cathode, consisting of an electronic and ionic conducting phase, the electro-chemical active three-phase-boundary can be extended into the volume of the cathode.

**anode material:** Ni-YSZ (ceramic-metal) composite → Cermet

- Reducing  $H_2$  atmosphere and need for good el. conductivity → metals, e.g. Ni;
- problem: High CTE and high sintering tendency of most metals → thermally induced stresses, loss of contact
- solution: cermet anodes made of Ni and YSZ → Reduction of CTE, introduction of  $\sigma_{ion}$  and generation of porosity by volume shrinkage  $NiO \rightarrow Ni$ ; shaping from NiO and YSZ, reduction to Ni-YSZ cermet during 1st operating cycle of BZ



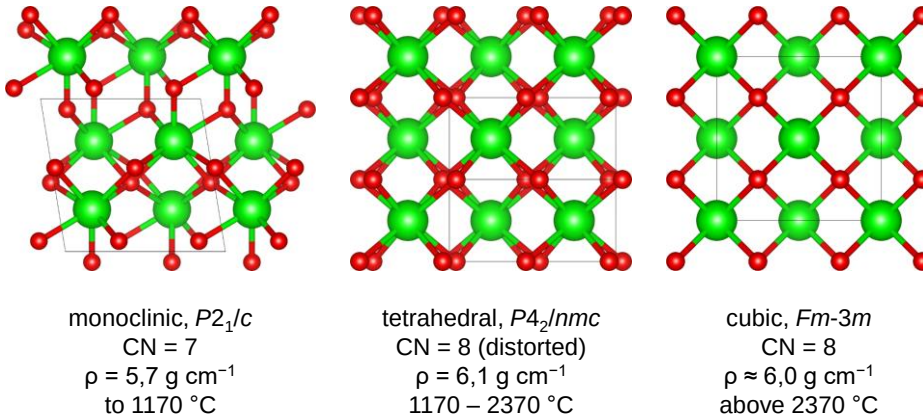
Transport processes at a Ni/YSZ-Cermet anode



in contrast to the metal oxide based cathode structures, electron or oxygen ion transport in a Ni/YSZ cermet anode are restricted to the metallic (Ni) or resp. ceramic (YSZ) phase; it is crucial that both phases form a continuous matrix; if some single Ni or YSZ particles have no electric contact to the interconnector or to the electrolyte substrate, they remain inactive

**elektrolyte:** YSZ –  $\text{Zr}_{1-x}\text{Y}_x\text{O}_{2-x/2}$   $x \approx 0.16$

ionic compound → localized electrons → poor electronic conductor  $\text{ZrO}_2$  is polymorphic, from RT-1170 °C the monoclinic form is stable, but at sintering temperature tetragonal  $\text{ZrO}_2$  → phase transition with volume change

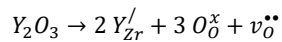


**elektrolyte:** YSZ –  $\text{Zr}_{1-x}\text{Y}_x\text{O}_{2-x/2}$   $x \approx 0.16$

ionic compound → localized electrons → poor electronic conductor  $\text{ZrO}_2$  is polymorphic, from RT-1170 °C the monoclinic form is stable, but at sintering temperature tetragonal  $\text{ZrO}_2$  → phase transition with volume change

→ stabilization of the cubic phase (fluorite type) at RT by  $\text{Y}_2\text{O}_3$  doping

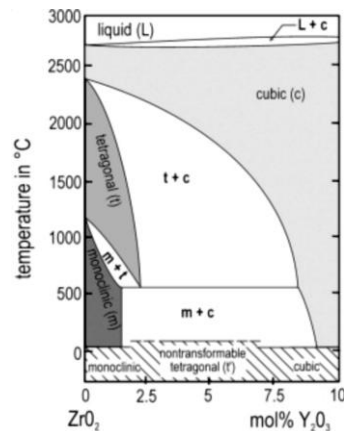
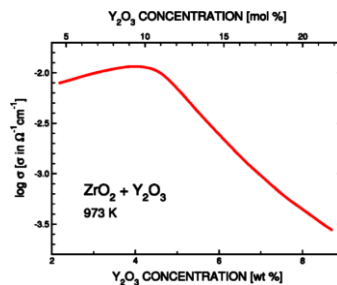
→ Generation of oxygen vacancies:



→ conductivity f.  $\text{O}^{2-}$ , max. at  $\approx 10 \text{ mol } \% \text{ Y}_2\text{O}_3$

higher concentrations:

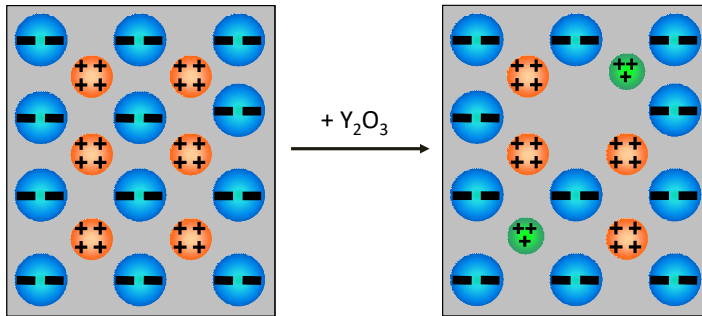
clustering of  $v_\text{O}^{\bullet\bullet}$



## repeation

What happens in the crystal structure when  $\text{Zr}^{4+}$  is partially substituted by  $\text{Y}^{3+}$ ?

doping of  $\text{ZrO}_2$  with  $\text{Y}_2\text{O}_3$

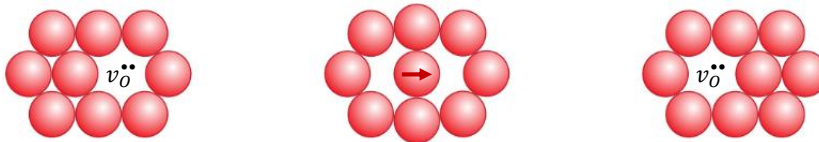


$$r_{\text{Zr}^{4+}} = 98 \text{ pm}$$

$$r_{\text{Y}^{3+}} = 116 \text{ pm}$$

- massive distortion of the crystal lattice
- t → m transformation does not occur under ambient conditions
- defects formed...useable for ion conductivity

**elektrolyte:** YSZ –  $\text{Zr}_{1-x}\text{Y}_x\text{O}_{2-x/2}$   $x \approx 0.16$ ; conductivity by vacancy hopping



ion conductivity depends on the number of  $n_{v_O^{\bullet\bullet}}$  of the charge carriers ( $v_O^{\bullet\bullet}$ ) and their mobility  $\mu_{v_O^{\bullet\bullet}}$ :

$$\sigma(T) = 2e_0 \cdot \mu_{v_O^{\bullet\bullet}}(T) \cdot n_{v_O^{\bullet\bullet}}$$

an Arrhenius-like relationship applies to the temperature dependence of mobility:

$$\mu_{v_O^{\bullet\bullet}}(T) = \mu_{v_O^{\bullet\bullet}}^0 \cdot \exp\left(\frac{-E_A}{kT}\right) \rightarrow$$

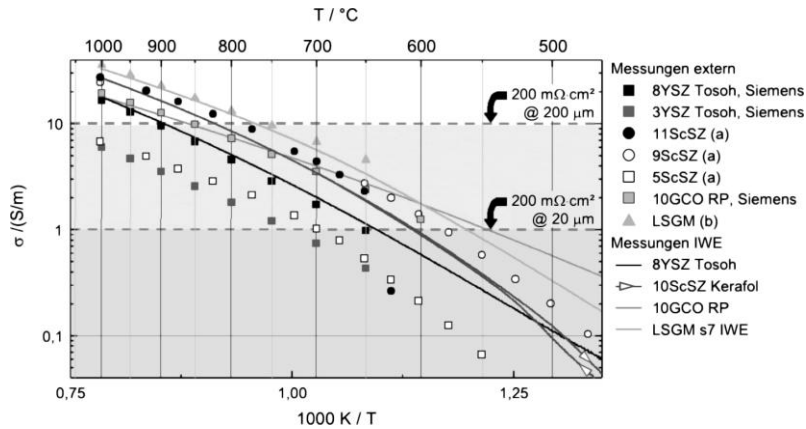
$$\sigma(T) = 2e_0 \cdot \mu_{v_O^{\bullet\bullet}}^0 \cdot \exp\left(\frac{-E_A}{kT}\right) \cdot n_{v_O^{\bullet\bullet}}$$

charge carrier mobility and conductivity increase exponentially with T  
significant ionic conductivity for YSZ only above 800 °C ( $\approx 10 \text{ S m}^{-1}$ )  
main reason for the high operating temperatures of SOFCs

**elektrolyte:** YSZ –  $\text{Zr}_{1-x}\text{Y}_x\text{O}_{2-x/2}$   $x \approx 0.16$

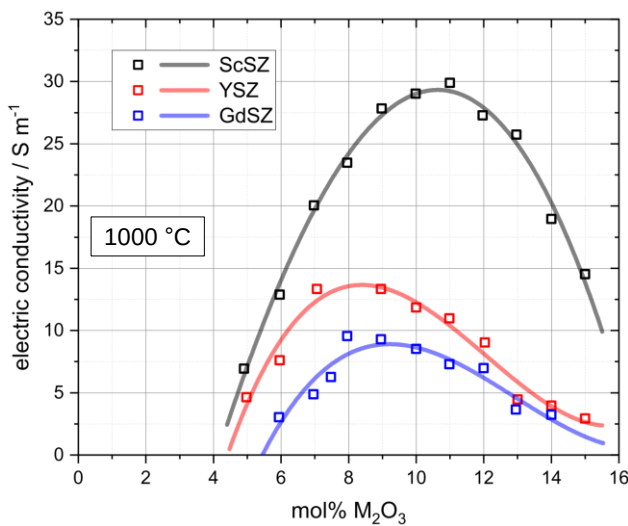
Ion conductivity depends on the number  $n_{v_o^{\bullet\bullet}}$  of the charge carriers ( $v_o^{\bullet\bullet}$ ) und deren Beweglichkeit and their mobility  $\mu_{v_o^{\bullet\bullet}}$ :

$$\sigma(T) = 2e_0 \cdot \mu_{v_o^{\bullet\bullet}}^0 \cdot \exp\left(\frac{-E_A}{kT}\right) \cdot n_{v_o^{\bullet\bullet}}$$



**elektrolyte:** different materials

doping of  $\text{ZrO}_2$  with RE oxides  $\text{RE}_2\text{O}_3$  (SE = Sc, Y, Gd)



Ion radii (CN = 8):

|                  |        |
|------------------|--------|
| $\text{Zr}^{4+}$ | 98 pm  |
| $\text{Sc}^{3+}$ | 101 pm |
| $\text{Y}^{3+}$  | 116 pm |
| $\text{Gd}^{3+}$ | 119 pm |

the more the radius of the dopant ion deviates from that of  $\text{Zr}^{4+}$ , the stronger the distortion of the lattice and the lower the conductivity  
ScSZ optimal, but  $\text{Sc}_2\text{O}_3$  is very expensive